DS701 Homework 8: SVD and PCA

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SOLUTIONS

Problem 1: Reading a singular-value spectrum

A building has 300 temperature sensors, read hourly for 2,000 hours: $A \in \mathbb{R}^{2000\times 300}$ (rows = hours, columns = sensors). Its SVD $A = U\Sigma V^T$ has singular values $\sigma_1 = 40$, $\sigma_2 = 30$, $\sigma_3 = 10$, $\sigma_4 = 8$, $\sigma_5 = 6$, and $\sigma_6, \dots, \sigma_{300}$ all below 10^{-12}.

(a) What is the rank of A? One sentence on why the tail of tiny values does not count, and one on what this rank says about the 300 sensors (think of $A = CR$).

rank $A =$ _____

(b) From the spectrum alone, fill in the blanks for $k = 1, 2, 3$. Then: what is the smallest k whose $A^{(k)}$ captures at least 95% of $\|A\|_F^2$, and for that k, how many numbers does the factored $A^{(k)}$ store, as a fraction of mn?

Solution

(a) rank $A = 5$. The rank is the number of **nonzero** singular values; $\sigma_6, \dots, \sigma_{300} \approx 10^{-12}$ are floating-point zeros (round-off from computing the SVD), not genuine structure, so they do not count. Rank 5 means every one of the 300 sensor columns is a linear combination of just 5 underlying signals: $A = CR$ with $C \in \mathbb{R}^{2000\times 5}$ (five hidden temperature signals over time) and $R \in \mathbb{R}^{5\times 300}$ (how each sensor mixes them). *Grading:* full credit for "5" plus either sentence; a student who answers 300 ("the matrix is full rank") has confused the reported values with exact zeros.

(b) $\|A\|_F^2 = \sum_i \sigma_i^2 = 1600+900+100+64+36 = 2700$. The captured fraction is $\sum_{i\le k} \sigma_i^2/2700$, the error is the tail $\|A - A^{(k)}\|_F^2 = \sum_{i>k} \sigma_i^2$, and factored storage is $k(m+n) = 2300k$:

	$k=1$	$k=2$	$k=3$
fraction captured	$1600/2700 = 0.593$	$2500/2700 = 0.926$	$2600/2700 = 0.963$
$\|A - A^{(k)}\|_F^2$	1100	200	100

$k = 2$ reaches 92.6%, $k = 3$ reaches 96.3%, so $k = 3$. Storage for $k = 3$: $k(m+n) = 3{\cdot}2300 = 6{,}900$ numbers versus $mn = 600{,}000$, a fraction $6{,}900/600{,}000 = 0.0115$, about 1.2% of A. *Common slips:* using σ_i instead of σ_i^2 (fractions $40/94 = 0.43$, …) — the sheet's Frobenius identity is in squares; reporting the error as $\sqrt{200} \approx 14.1$ when the squared norm was asked.

Problem 2: What the singular vectors mean

Practices: reading the first two singular vectors of a small table as patterns / latent factors. Background: Appendix 12B (Reading the Singular Vectors) works a similar example step by step.

A café records how many of four products five regulars bought last month:

	coffee	filters	tea	honey
customer 1	8	6	1	0
customer 2	6	5	0	1
customer 3	1	0	7	6
customer 4	0	1	8	7
customer 5	4	3	4	3

Its SVD $A = U\Sigma V^T$ has $\sigma = (16.1,\ 12.3,\ 1.2,\ 0.8)$, and the first two singular-vector pairs are (rounded to two decimals; signs chosen so that $\mathbf{v}_1$ is positive and the first entry of $\mathbf{v}_2$ is positive):

$$\mathbf{v}_1 = \begin{bmatrix} 0.45 \\ 0.36 \\ 0.63 \\ 0.53 \end{bmatrix} \begin{matrix} \text{coffee} \\ \text{filters} \\ \text{tea} \\ \text{honey} \end{matrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0.65 \\ 0.50 \\ -0.43 \\ -0.38 \end{bmatrix}, \qquad \mathbf{u}_1 = \begin{bmatrix} 0.40 \\ 0.31 \\ 0.50 \\ 0.56 \\ 0.43 \end{bmatrix} \begin{matrix} \text{c1} \\ \text{c2} \\ \text{c3}, \\ \text{c4} \\ \text{c5} \end{matrix} \qquad \mathbf{u}_2 = \begin{bmatrix} 0.63 \\ 0.49 \\ -0.38 \\ -0.46 \\ 0.10 \end{bmatrix}.$$

(a) In one sentence, what pattern in the products does $\mathbf{v}_1$ describe, and what does a customer's entry in $\mathbf{u}_1$ measure? Which customer has the largest first coordinate in $U'\Sigma'$ (the sheet's "row of $U'\Sigma'$")? Use the printed numbers — one multiplication at most.

Customer: _____

(b) Now the second pair. In one or two sentences, what *contrast* does $\mathbf{v}_2$ capture, and which customers does $\mathbf{u}_2$ put on each side of it? What does customer 5's entry 0.10 say about them? Finally: $\sigma_1^2 + \sigma_2^2$ is 99.5% of $\|A\|_F^2$ — in one sentence, what does that tell you about this table?

Solution

(a) $\mathbf{v}_1$ is positive on every product with similar weights, so it is the **overall-buying** pattern — a "typical basket" direction; the first coordinate of a customer's row of $U'\Sigma'$ is $\sigma_1 u_{i1}$, which measures **how much that customer buys overall** (their loading on the typical basket). The largest entry of $\mathbf{u}_1$ is customer 4's (0.56), so customer 4 has the largest first coordinate, $16.1 \times 0.56 \approx 9.0$ (customer 3 is next at $16.1 \times 0.50 \approx 8.0$). *Grading:* "overall volume / everyone buys a bit of everything" earns the pattern credit; customer 4 for the loading (no need to multiply — the ordering of $\mathbf{u}_1$ is enough, since $\sigma_1 > 0$ is common to

all rows).

(b) $\mathbf{v}_2$ is positive on coffee and filters and negative on tea and honey: it is the **coffee-drinker versus tea-drinker contrast**. $\mathbf{u}_2$ places customers 1 and 2 $(+0.63, +0.49)$ on the coffee side and customers 3 and 4 $(-0.38, -0.46)$ on the tea side; customer 5's $0.10 \approx 0$ means they sit on neither side — they buy both kinds in equal measure (row 5 of A is $(4, 3, 4, 3)$), so the contrast factor barely applies to them. Two singular values carrying 99.5% of $\|A\|_F^2$ means the table is essentially **rank 2**: two latent factors (overall volume and the coffee/tea preference) reproduce it to within a rounding error, and $A^{(2)}$ would be an excellent reconstruction. *Grading:* the sign reading of $\mathbf{v}_2$ and $\mathbf{u}_2$ is the point; any sensible name for the contrast is fine. For the last sentence, "two factors are enough / the effective rank is 2 / $A^{(2)} \approx A$" all earn full credit.

Problem 3: PCA variance accounting, preprocessing, and a t-SNE map

Practices: reading eigenvalues as variances, knowing why centering and scaling matter, and reading a t-SNE picture without over-reading it.

(a) A dataset has $m = 101$ rows and $n = 6$ features. The features are **standardized** (each column centered and scaled to variance 1) before PCA, and the covariance matrix S has eigenvalues

$$\lambda = (3.0,\ 1.5,\ 0.9,\ 0.3,\ 0.2,\ 0.1).$$

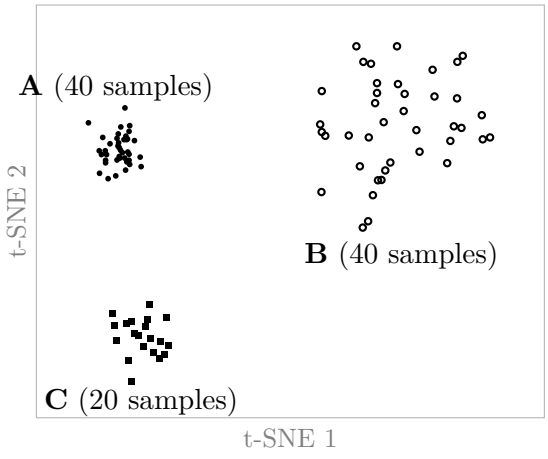
Fill in the table. What is the smallest number of components d that retains at least 90% of the variance? Why is the total T exactly 6 here, in one sentence? Lastly, if the centered, standardized data matrix X had been decomposed by SVD instead, what would σ_1 be?

(b) True or false, with one reason each.

(i) A teammate's features are house price (dollars), age (years) and number of rooms. They say: "I centered every column, so PCA is fine — scaling is optional."

(ii) Another teammate skips centering and takes the top right singular vector of the raw X_0 as "PC1": "the top singular vector is the direction of maximum variance either way."

(c) The figure is a t-SNE map (perplexity 30, one random start) of 100 tissue samples, each originally described by 50 PCA features. Three groups are visible.



For each claim, say **supported** or **not supported** by this map, with one reason.

(i) "Group B's samples are more varied among themselves than group A's - look how much more spread out B is."

(ii) "Group A is biologically closer to group C than to group B, since A sits nearer to C."

(iii) "Two samples that sit next to each other inside group A are near neighbours in the original 50-dimensional space."

(iv) "Re-running t-SNE with a different random seed would reproduce this layout."

Finally: the team wants to train a classifier on a 20-number summary of each sample. Which method, PCA or t-SNE, should produce those 20 numbers, and why (one sentence)?

Solution

(a) $T = 3.0 + 1.5 + 0.9 + 0.3 + 0.2 + 0.1 = 6.0$.

PC i	λ_i	ratio λ_i/T	cumulative
1	3.0	0.50	0.50
2	1.5	0.25	0.75
3	0.9	0.15	0.90
4	0.3	0.05	0.95
5	0.2	0.033	0.983
6	0.1	0.017	1.00

Two components give 75%, three give exactly 90%, so $d = 3$. $T = \operatorname{tr} S$ is the sum of the feature variances, and every standardized feature has variance 1, so $T = n = 6$. Via SVD, $\lambda_1 = \sigma_1^2/(m-1)$, so $\sigma_1 = \sqrt{\lambda_1(m-1)} = \sqrt{3 \cdot 100} = \sqrt{300} = 10\sqrt{3} \approx 17.3$. *Grading:* the table and d are the core; the $T = 6$ sentence needs "variance 1 per standardized column"; for σ_1 accept $\sqrt{300}$ in any form. A common slip is $\sigma_1 = \sqrt{3 \cdot 101}$ — the sheet's denominator is $m - 1$.

(b) (i) **False.** Centering fixes the mean but not the scale: price in dollars has a variance millions of times larger than age or rooms, so PC1 would simply be the price axis. With features in different units, standardize (divide each column by its standard deviation) as well — the sheet's step (1).

(ii) **False.** $\frac{1}{m-1}X^TX$ is the covariance matrix only when the columns have mean zero. For uncentered data the top right singular vector points from the origin toward the data's **mean** (the direction of largest average squared coordinate), not along the direction of largest spread, so the "PC1" and all the variance accounting in (a) would be wrong. *Grading:* each needs the verdict and the specific reason (dominant scale; points at the mean). "False because the sheet says to standardize" is half credit.

(iii) (i) **Not supported.** t-SNE does not preserve cluster sizes or spreads; how tight or loose a group looks is an artifact of the embedding (and the perplexity), not a measure of within-group variability.

(iv) **Not supported.** Distances *between* clusters in a t-SNE map are not meaningful; only local neighbourhoods are. A sitting nearer C than B says nothing about which groups are more similar.

(v) **Supported** (with the usual caveat). t-SNE is built to preserve local neighbourhoods — the high-dimensional similarities p_{ij} of nearby points — so points that land next to each other were, with high probability, near neighbours in the 50-D PCA space. That is the one thing the map is reliable about.

(vi) **Not supported.** t-SNE starts from random low-dimensional positions and optimizes a non-convex objective, so a different seed gives a different layout (groups may move, rotate or split); the sheet says to run it several times. **PCA** should produce the 20 features: it is a deterministic linear transform ($Y = XV'$) that can be applied to new samples and preserves global variance structure, whereas t-SNE is a visualization tool with no transform for new points and distances that mean nothing to a downstream model. *Grading:* 4 verdicts with the right reason type (size / inter-cluster distance / local neighbourhood / random init), plus PCA with "reusable / applies to new data" or "t-SNE is visualization only". The map's actual geometry is deliberately misleading (B is spread, A is nearer C) so that reading it naively gives the wrong answer.